Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Enterprise Java (EJ)

Practical – III

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Date of Assignment: 08/07/2026	Date/Time of Submission: 08/07/2026

AIM: <u>Implement the Servlet IO and File applications.</u>

A. Create a Servlet application to upload and download a file.

CODE:

(index.html)

```
<!DOCTYPE html>
<html>
<head>
<title>File Upload</title>
</head>
<body>
<h1>File Upload Application</h1>
<form action="FileUploadServlet" method="post" enctype="multipart/form-data">
Select file to upload: <input type="file" name="file" required>
<br><br>
<input type="submit" value="Upload File">
</form>
<h2>Download Links</h2>
<!-- Link to download files, ensure the path is correct -->
<a href="FileDownloadServlet?filename=Enterprise JAVA Practical.pdf">Download the uploaded
file</a><br><br><br><br>
</body>
</html>
```

(FileUploadServlet.java)

```
import java.io.File;
import java.io.FileOutputStream;
import java.io.IOException;
import java.io.InputStream;
import java.io.OutputStream;

import javax.servlet.ServletException;
import javax.servlet.annotation.MultipartConfig;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
```

```java
import javax.servlet.http.Part;
@WebServlet("/FileUploadServlet")
@MultipartConfig
public class FileUploadServlet extends HttpServlet {
private static final String UPLOAD_DIR = "uploads";
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
String uploadPath = getServletContext().getRealPath("") + File.separator + UPLOAD_DIR;
File uploadDir = new File(uploadPath);
if (!uploadDir.exists()) {
uploadDir.mkdir();
}
Part filePart = request.getPart("file");
String fileName = filePart.getSubmittedFileName();
File file = new File(uploadPath + File.separator + fileName);
try (InputStream inputStream = filePart.getInputStream();
OutputStream outputStream = new FileOutputStream(file)) {
byte[] buffer = new byte[4096];
int bytesRead;
while ((bytesRead = inputStream.read(buffer)) != -1) {
outputStream.write(buffer, 0, bytesRead);

}
}
response.getWriter().println("File uploaded successfully.");
}
}
```

(FileDownloadServlet.java)

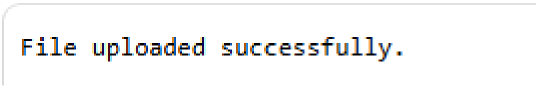
```java
import java.io.File;
import java.io.FileInputStream;
import java.io.IOException;
import java.io.OutputStream;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet("/FileDownloadServlet")
public class FileDownloadServlet extends HttpServlet {
private static final String UPLOAD_DIR = "uploads";
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
String fileName = request.getParameter("filename");
if (fileName == null) {
response.getWriter().println("File name is missing.");
```

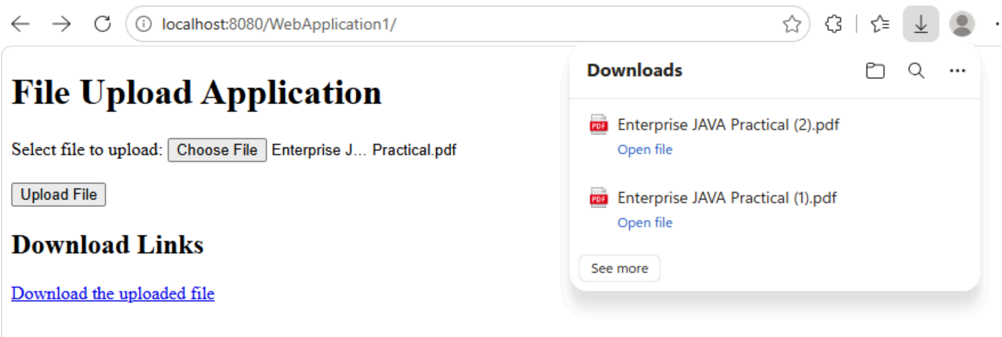
```
return;
}
String filePath = getServletContext().getRealPath("") + File.separator + UPLOAD_DIR +
File.separator + fileName;

File file = new File(filePath);
if (!file.exists()) {
response.getWriter().println("File not found.");
return;
}
response.setContentType("application/octet-stream");
response.setHeader("Content-Disposition", "attachment; filename=\"" + fileName +
"\"");
try (FileInputStream inputStream = new FileInputStream(file);
OutputStream outputStream = response.getOutputStream()) {
byte[] buffer = new byte[4096];
int bytesRead;
while ((bytesRead = inputStream.read(buffer)) != -1) {
outputStream.write(buffer, 0, bytesRead);
}}}}
```

(web.xml)

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE glassfish-web-app PUBLIC "-//GlassFish.org//DTD GlassFish Application Server
3.1 Servlet 3.0//EN" "http://glassfish.org/dtds/glassfish-web-app_3_0-1.dtd">
<glassfish-web-app error-url="">
<context-root>/FileUploadanDownload</context-root>
<class-loader delegate="true"/>
<jsp-config>
<property name="keepgenerated" value="true">
<description>Keep a copy of the generated servlet class' java code.</description>
</property>
</jsp-config>
</glassfish-web-app>
```

OUTPUT:





AIM(B): - Develop Simple Servlet Question Answer Application using Database.

CODE B :

```
HTML :  <!DOCTYPE html>
<html>
<head>
<title>Java Quiz</title>
</head>
<body bgcolor="lightyellow">
<center>
<h1>Java Quiz Application</h1>
<h3>Click below to start the quiz</h3>
<a href="QuizServlet">START QUIZ</a>
</center>
</body>
</html>
```

Servlet :

QuizServlet

```
package mypack;

import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class QuizServlet extends HttpServlet {

    public void doGet(HttpServletRequest request,
                      HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<html>");
```

```java
        out.println("<body bgcolor='lightblue'>");
        out.println("<center>");
        out.println("<h1>Java Quiz</h1>");
        out.println("</center>");

        out.println("<form action='ShowResult' method='get'>");

        try {

            Class.forName("com.mysql.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/qadb",
                    "root",
                    "root");

            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("SELECT * FROM quiz");

            int i = 1;

            while (rs.next()) {

                out.println("<hr>");
                out.println("<h3>Q" + i + ". " + rs.getString("question") + "</h3>");

                out.println("<input type='radio' name='" + i + "' value='" + rs.getString("op1") + "'>" +
rs.getString("op1") + "<br>");

                out.println("<input type='radio' name='" + i + "' value='" + rs.getString("op2") + "'>" +
rs.getString("op2") + "<br>");

                out.println("<input type='radio' name='" + i + "' value='" + rs.getString("op3") + "'>" +
rs.getString("op3") + "<br>");

                out.println("<input type='radio' name='" + i + "' value='" + rs.getString("op4") + "'>" +
rs.getString("op4") + "<br>");

                i++;
            }

            rs.close();
            st.close();
            con.close();

        } catch (Exception e) {
            out.println("<h3>Error: " + e.getMessage() + "</h3>");
        }

        out.println("<br><br>");
        out.println("<input type='submit' value='Submit Quiz'>");
        out.println("<input type='reset' value='Reset'>");
        out.println("</form>");
        out.println("</body>");
        out.println("</html>");
```

```
    }
}
```

ShowResult.java:

```
package mypack;

import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class ShowResult extends HttpServlet {

    public void doGet(HttpServletRequest request,
            HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<html>");
        out.println("<body bgcolor='lightgreen'>");
        out.println("<center>");
        out.println("<h1>Quiz Result</h1>");
        out.println("</center>");

        try {

            Class.forName("com.mysql.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/qadb",
                    "root",
                    "root");

            Statement st = con.createStatement();

            ResultSet rs = st.executeQuery("SELECT ans FROM quiz");

            int score = 0;
            int i = 1;

            while (rs.next()) {

                String correct = rs.getString("ans");
                String user = request.getParameter(String.valueOf(i));

                if (user != null && correct.equals(user)) {
                    score++;
                    out.println("<h3>Q" + i + " : Correct</h3>");
                } else {
                    out.println("<h3>Q" + i + " : Incorrect</h3>");
                }

                i++;
```

```
            }

            out.println("<hr>");
            out.println("<h2>Your Score : " + score + " / " + (i - 1) + "</h2>");

            rs.close();
            st.close();
            con.close();

        } catch (Exception e) {
            out.println("<h3>Error : " + e.getMessage() + "</h3>");
        }

        out.println("<br><br>");
        out.println("<a href='index.html'>Take Quiz Again</a>");
        out.println("</body>");
        out.println("</html>");
    }
}
```

web.xml :

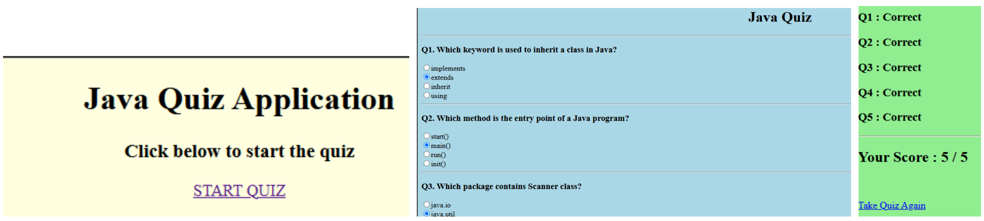
```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
http://xmlns.jcp.org/xml/ns/javaee/web-app_3_1.xsd"
version="3.1">
<servlet>
<servlet-name>QuizServlet</servlet-name>
<servlet-class>mypack.QuizServlet</servlet-class>
</servlet>
<servlet-mapping>
<servlet-name>QuizServlet</servlet-name>
<url-pattern>/QuizServlet</url-pattern>
</servlet-mapping>
<servlet>
<servlet-name>ShowResult</servlet-name>
<servlet-class>mypack.ShowResult</servlet-class>
</servlet>
<servlet-mapping>
<servlet-name>ShowResult</servlet-name>
<url-pattern>/ShowResult</url-pattern>
</servlet-mapping>
</web-app>
```

SQL CODE :

```
mysql> CREATE DATABASE qadb;
Query OK, 1 row affected (0.00 sec)

mysql> USE qadb;
Database changed
mysql> CREATE TABLE quiz (
    ->     qno VARCHAR(5) PRIMARY KEY,
    ->     question VARCHAR(200),
    ->     op1 VARCHAR(100),
```

```
    ->     op2 VARCHAR(100),
    ->     op3 VARCHAR(100),
    ->     op4 VARCHAR(100),
    ->     ans VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO quiz VALUES
    -> ('001','Which keyword is used to inherit a class in Java?','implements','extends','inherit','using','extends');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO quiz VALUES
    -> ('002','Which method is the entry point of a Java program?','start()','main()','run()','init()','main()');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO quiz VALUES
    -> ('003','Which package contains Scanner class?','java.io','java.util','java.lang','java.sql','java.util');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO quiz VALUES
    -> ('004','Which is not a primitive data type?','int','float','String','char','String');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO quiz VALUES
    -> ('005','Which exception occurs when a number is divided by zero?','IOException','NullPointerException','ArithmeticException','ClassNotFoundException','ArithmeticException');
Query OK, 1 row affected (0.01 sec)
```

OUTPUT :



AIM(C): Create simple Servlet application to demonstrate Non-Blocking Read Operation.

CODE :

```
HTML :<!DOCTYPE html>
<html>
<head>
<title>Non-Blocking Practical</title>
</head>
<body>
<form action="NBServ" method="post">
<h3>Please Click Start Button to Start Non-Blocking Read!!!</h3>
<input type="submit" value="Start NB IO Read">
</form>
<div>TO DO write content</div>
</body>
</html>
```

Servlet :

MyReadListener.java :

```
package EJPracts;
import java.io.IOException;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.servlet.AsyncContext;
import javax.servlet.ReadListener;
import javax.servlet.ServletInputStream;
public class MyReadListener implements ReadListener {
private ServletInputStream iStream = null;
private AsyncContext cnt = null;
public MyReadListener(ServletInputStream in, AsyncContext ac) {
this.iStream = in;
this.cnt = ac;
}
@Override
public void onDataAvailable() {
try {
StringBuilder sb = new StringBuilder();
int len;
byte[] b = new byte[1024];
while (iStream.isReady() && (len = iStream.read(b)) != -1) {
String mydata = new String(b, 0, len);
sb.append(mydata);
}
} catch (IOException ex) {
Logger.getLogger(MyReadListener.class.getName()).log(Level.SEVERE, null, ex);
}
}

@Override
public void onAllDataRead() {
System.out.println("Called onAllDataRead()");
cnt.complete();
}
@Override
public void onError(Throwable t) {
Logger.getLogger(MyReadListener.class.getName()).log(Level.SEVERE, null, t);
cnt.complete();
}
}
```

NBServ.java

```
import java.io.BufferedReader;
import java.io.BufferedWriter;
import java.io.IOException;
import java.io.InputStream;
import java.io.InputStreamReader;
import java.io.OutputStreamWriter;
import java.io.PrintWriter;
import java.net.HttpURLConnection;
import java.net.URL;
import java.util.logging.Level;
import java.util.logging.Logger;
```

```
import javax.servlet.ServletContext;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet(urlPatterns = {"/NBServ"})
public class NBServ extends HttpServlet {
protected void processRequest(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
response.setContentType("text/html;charset=UTF-8");
String filepath = "/WEB-INF/Harshita.txt";
ServletContext cnt = getServletContext();
InputStream iStream = cnt.getResourceAsStream(filepath);
try (PrintWriter out = response.getWriter()) {
String path = "http://" + request.getServerName() + ":" + request.getServerPort() +
request.getContextPath() + "/ReadNBServ";
out.println("<html>");
out.println("<head>");
out.println("<title>Non-Blocking I/O Servlet</title>");
out.println("</head>");
out.println("<body>");
out.println("<h1>Reading Harshita Kanojia file</h1>");
out.flush();

URL url = new URL(path);
HttpURLConnection mycon = (HttpURLConnection) url.openConnection();
mycon.setChunkedStreamingMode(2);
mycon.setDoOutput(true);
mycon.connect();
if (iStream != null) {
InputStreamReader isr = new InputStreamReader(iStream);
BufferedReader br = new BufferedReader(isr);
String readText;
try (BufferedWriter bw = new BufferedWriter(new
OutputStreamWriter(mycon.getOutputStream()))) {
out.println("<div style='width=100%;height:450px;overflow:scroll;'>");
while ((readText = br.readLine()) != null) {
out.println("<div style='background-color:khaki;width=100%;'>");
out.println(readText);
out.println("</div><br/>");
out.flush();
bw.write(readText);
Thread.sleep(1000);
out.flush();
}
out.println("</div>");
bw.flush();
}
}
out.println("</body>");
out.println("</html>");
} catch (InterruptedException | IOException ex) {
Logger.getLogger(NBServ.class.getName()).log(Level.SEVERE, null, ex);
}
```

```
}
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
processRequest(request, response);
}
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
processRequest(request, response);
}
@Override
public String getServletInfo() {
return "Short description";
}
}
```

ReadNBServ.java :

```
import EJPracts.MyReadListener;
import java.io.IOException;

import java.io.PrintWriter;
import javax.servlet.AsyncContext;
import javax.servlet.ServletException;
import javax.servlet.ServletInputStream;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet(name = "ReadNBServ", urlPatterns = {"/ReadNBServ"}, asyncSupported = true)
public class ReadNBServ extends HttpServlet {
protected void processRequest(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
response.setContentType("text/html;charset=UTF-8");
try (PrintWriter out = response.getWriter()) {
out.println("<!DOCTYPE html>");
out.println("<html>");
out.println("<head>");
out.println("<title>Servlet ReadNBServ</title>");
out.println("</head>");
out.println("<body>");
AsyncContext cnt = request.startAsync();
ServletInputStream iStream = request.getInputStream();
iStream.setReadListener(new MyReadListener(iStream, cnt));
out.println("</body>");
out.println("</html>");
}
}
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
processRequest(request, response);
}
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
```

```java
throws ServletException, IOException {
processRequest(request, response);
}
@Override
public String getServletInfo() {
return "Short description";
}
}
```

OUTPUT :